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REVIEW ARTICLE

Insights into the relationships between physicochemical properties, solvent performance, and applications of deep eutectic solvents

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Abstract

Deep eutectic solvent (DES) is regarded as a new generation of green solvent due to its distinctive and tailorable physicochemical properties, such as low volatility, strong solubility, biodegradability, low-cost, environment-friendly, and feasibility of the structural design. As an alternative to traditional organic solvents and ionic liquids (ILs), DESs have been widely applied in many fields, such as organic chemical synthesis, electrochemical deposition, material preparation, biomass catalytic conversion, extraction and separation, detection and analysis, nanotechnology, gas absorption, and drug delivery. In this paper, through in-depth discussion on factors influencing the physicochemical properties of DESs, we summarized the relations between their composition, structure, and performance. Focusing on their solvent performance, we analyzed the latest research results of DESs with different physicochemical properties in various fields. It should be pointed out that designing and synthesizing DESs from the molecular structure aspect to regulate their physicochemical properties is the direction of accurately developing new functional applications of DESs.

Keywords Deep eutectic solvent, · Green solvent, · Physicochemical properties, · Solvent performance, · Molecular structure, · Functional applications

Introduction

The solvent is an essential medium in the process of chemical reaction and separation. Green technology actively seeks new solvents with excellent solvent performance and environment-friendly merits to replace conventional organic solvents which are inherent toxicity and high volatility, thus reducing environmental pollution (El Achkar et al. 2019). Ionic liquids (ILs) have attracted considerable attention in the research and industrial fields because of their high thermal stability, strong solvation ability, and low vapor pressure. Nevertheless, the "greenness" of ILs is often questioned, mainly due to their complex synthesis process, difficult purification, high toxicity, low synthesis atom economy, and poor biodegradability (Kostić and Divac 2019).

More and more attention has focused on the deep eutectic solvents (DESs) which are analogous but superior to ILs. DESs have similar excellent physicochemical properties and solvent performance with ILs but are cheaper to produce and are often biodegradable (Moura et al. 2017). Thus, they are foreseen as an alternative to ILs. The DES is composed of a fixed stoichiometric ratio of hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) which forms a binary or ternary mixed system, and its freezing point is lower than that of each component (Abbott et al. 2006b; Zhang et al. 2012). Abbott et al. prepared DESs for the first time in 2003 by mixing two solid substances with high freezing points (choline chloride (ChCl) and urea) (Abbott et al. 2003). Jiang et al. synthesized a class of ternary DESs by mixing ChCl, polyethylene glycol, and boric acid. Through tuning the molar ratio of the three compositions, the desulfurization efficiency can be optimized. Besides, compared with ChCl-organic acid binary DESs, those ternary DESs offer advantages including low volatility, low toxicity, and high activity (Jiang et al. 2019). Inspired by this work, ternary DESs may provide more ideas for designing task-specific DESs. When the components of

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HBA and HBD are metabolites from living organisms, such as choline derivatives, alcohols, sugars, urea, organic acids, and amino acids, they are also named as natural deep eutectic solvents (NADESs) (Tang et al. 2017). NADESs are the new type of solvents concept proposed by Choi et al. in 2011, that is, viscous liquid solvents formed entirely from small molecular metabolites of organisms. They fully represent the principle of green chemistry and are considered as safer solvents (Choi et al. 2011).

Similar to ILs, DESs are also a novel class of designable solvents (Krishnan et al. 2020). By modifying the composition, molar ratio, or functional group, it is possible to prepare DESs with a wide range of physicochemical properties, such as polarity, conductivity, acidity/alkalinity, and hydrophilicity/hydrophobicity. A large number of possible DESs combinations can also adjust the solvent performance to adapt to their much different applications (Patzold et al. 2019). In addition, temperature and water content may also affect the physicochemical properties of DESs. For example, increasing temperature and water content will reduce the viscosity and density of the systems.

DESs have low vapor pressure, high thermal stability, strong solvation ability, wide electrochemical potential energy window, and the tunable characteristic of their properties, which provide a new choice for industrial production under mild conditions (Francisco et al. 2012). Due to the high thermal stability and low vapor pressure, DESs can participate in reactions above 100 °C in nonpressurized vessels. Consequently, DESs have received considerable attention in the fields of chemical synthesis and biomass catalytic conversion as the eco-friendly and renewable solvents (Vanda et al. 2018; Xu et al. 2017). As the potential alternative medium for organic reactions, DESs not only reduce the burden of solvents separation but also improve the efficiency of many organic reactions and reduce the pollution of volatile substances to the environment (Tomé et al. 2018; Chang 2020). DESs are used as electrolytes in electrodeposition and other electrochemical applications due to their wide electrochemical window and good electrical conductivity (Smith 2013). They also show an extremely strong capacity to dissolve macromolecular substances, such as DNA, lignin, and starch (Vanda et al. 2018). The ability of DESs in solubilizing substances is relevant to their supramolecular structure and wide polarity range. They have displayed their versatility on lignin dissolution, natural products extraction, biodiesel purification, and drugs transportation (Dai et al. 2013b; Liu et al. 2018a; Xu et al. 2020). Furthermore, due to the ionic characteristics and high polarity of DESs, they also have a strong gas absorption capacity, such as the absorption of CO_2, SO_2, and H_2S (Hanif et al. 2020). Therefore, they can be applied in gas absorption, air purification, gas desulfurization, etc. (Elhamarnah et al. 2020).

DESs have strong intermolecular hydrogen bond interactions. The interaction and construction of each component largely determine the physical and chemical properties of solvents (such as boiling point, density, viscosity, polarity, and acidity/alkalinity, etc.). These properties can be designed by changing the composition and structure of HBD or HBA and their ratios to receive excellent applications in organic reactions, electrochemistry, biomass catalytic conversion, biodiesel production, extraction and separation, and gas absorption. As a consequence, it is necessary to know the relationships between the composition, structure, performance, and applications of DESs when designing and synthesizing the solvents applied in specific applications. Only by deeply understanding the "structure-function" relationship of the DESs systems and the influence of the structure and properties of DESs on industrial applications can DESs be better applied.

Classification of DESs

From the composition, DESs are categorized into four main types. Abbott and co-workers defined DESs using the general formula Cat^+X^-zY, where Cat^+, any type of phosphine, ammonium, and sulfonium cation; X, a Lewis base, generally a halide anion; Y, a certain Lewis or Bronsted acid; and z is the ratio of Y to the anion X. Table 1 shows the four categories of DESs (Loow et al. 2017). Type 1 of DESs is non-hydrated metal halides as Lewis acid. This type belongs to the same family as the more studied ILs of metal chlorides/imidazole, such as $FeCl_2$-[Bmim]Cl for NO absorption (Sun et al. 2018). Due to the high freezing points of Type I, the potential applicability is restricted in many fields. Type II of DESs is hydrous metal chloride salts, which is more feasible in industrial processes because of their inherent insensitivity to air and moisture (Smith et al. 2014). In type III of DES, the HBA constituents mostly include quaternary ammonium halide, phosphonium halide, and imidazolium halide, etc. A huge array of naturally existing HBD mainly contains amides, carboxylic acids, polyols, and sugars, etc. (Tan et al. 2020). The majority of HBA and HBD are solid components at room temperature (298 K), except for water, ethylene glycol, and glycerol. Type III is the most frequently studied category because of its rapid, simple preparation, non-reaction with water, biodegradability, and cost-effectiveness. Figure 1 summarizes the typical species of HBAs and HBDs used in common type III. Finally, Type IV of DESs is synthesized from metal salt and HBDs such as urea or polyol, providing a method to achieve liquefaction of metal salt without heating, thus broadening the temperature range of metal salts solvents. So far, type III is preferred over other types of DESs due to the availability of ingredients and environmental friendliness.

Structurally, DESs are divided into ionic and nonionic types. Most of the current studies are ionic DESs represented

Table 1 General formula for the classification of DESs (Loow et al. 2017)

I	Metal salt organic salt	Cat^+X^-zMClx; M=Zn, Sn, Fe, Al, Ga, In	$ZnCl_2$+ChCl
II	Metal salt hydrate+organic salt	$Cat^+X^-zMCl_x{\cdot}yH_2O$; M=Cr, Co, Cu, Ni, Fe	$CoCl_2{\cdot}6H_2O$+ ChCl
III	HBD+organic salt	Cat^+X^-zRZ; Z=$CONH_2$, COOH, OH	Urea+ChCl
IV	Zinc /aluminium chloride+HBD	$MCl_x + RZ=MCl_{x-1}{}^+{\cdot}RZ+MCl_{x+1}{}^-$; M=Al, Zn & Z=$CONH_2$, OH	$ZnCl_2$+urea

Cat+, any type of phosphine, ammonium, and sulfonium cation; X, a Lewis base; Y, a certain Lewis or Bronsted acid; and z is the ratio of Y to the anion X

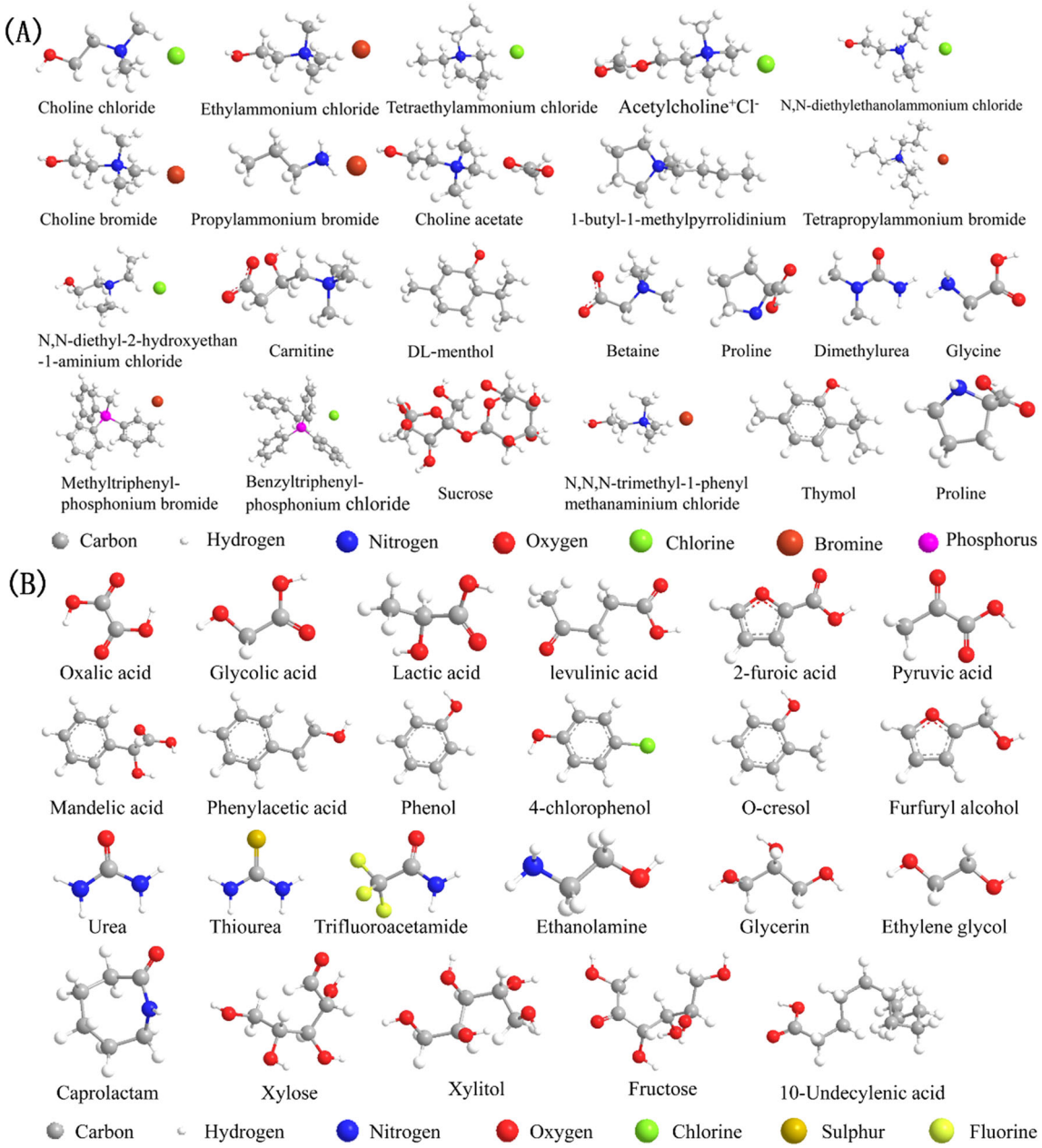


Fig. 1 Common HBA (**A**) and HBD (**B**) used for DESs synthesis

by ChCl as the HBA (Liu et al. 2018b). Due to the asymmetric, more massive ion structure and strong hydrogen bonding network, they are widely used in organic synthesis, biomass catalytic conversion, and electrochemistry (Paiva et al. 2014; Tang et al. 2017; Nguyen et al. 2017). Many nonionic DESs are composed of naturally originated components, thus belong to NADESs. The HBA components of nonionic DESs mainly include the following three types: sugars, amides mixtures, and zwitterionic amino acids, such as fructose, glucose, acetamide, and betaine (Zhou and Liu 2014; Van Osch et al. 2017; Dai et al. 2013a). And the HBD components of nonionic DESs are mostly carboxylic acids and amino acids (Zhou and Liu 2014). Since NADESs are composed of environmentally friendly ingredients, they are safer to use in edible production processes such as food or pharmaceuticals (Dai et al. 2014).

Physicochemical properties and applications

Freezing point and volatility

Some DESs and their components' freezing points are listed in Table 2. As shown in Table 2, freezing points of DESs are much lower than those of their HBA and HBD, and this property is attributed to large ion volume, asymmetric charge distribution, and the reduction of the coulomb force of DESs (Tang et al. 2015). The liquid state of DES is formed by lowering the freezing point, which can be measured by differential scanning calorimetry (DSC). It is worth noting that many nonionic DESs only show the glass transition temperatures (T_g) rather than freezing points, so they are also named low transition temperature mixtures (LTTM) (Francisco et al. 2012). For example, T_g of glucose with citric acid and tartaric acid at a molar ratio of 1:1 are 9.8 and −18.3 °C, respectively (Duan et al. 2016). As shown in Table 2, the freezing points or T_g of most DESs are within the range of −85 to 95 °C. If some DESs are solid, increasing the temperature or adding water could provide a wide range for DESs as the liquid solvents (Fig. 2).

The freezing points and liquid temperature ranges of DESs are mainly affected by ionic symmetry, alkane chain length, ability to form hydrogen bonds, etc. (Hou et al. 2018). With the loss of the ionic symmetry structure, the lower freezing temperature follows. For example, compared with other quaternary ammonium salts, ChCl-based DESs have lower lattice energy due to their larger asymmetric ions, which cause greater charge delocalization, resulting in lower freezing points (Smith et al. 2014). Organic acids as HBDs with low molecular weights showed the greatest decrease of freezing point in ChCl-organic acid DESs (Zhang et al. 2012). The freezing points of ChCl-oxalic acid/malonic acid/succinic acid (1:1) are 34 °C, 10 °C, and 71 °C, respectively (Abbott et al. 2004). Comparing with the melting points of pure organic acids, the decrease order of the freezing point of these three DESs is: ChCl-oxalic acid (156°C)>malonic acid (125°C)>succinic acid (114°C). The shorter the carbon chain, the lower the molecular weight of the organic acid, so the greater the freezing point reduction of DES. Thus, the oxalic acid with the lowest length of alkyl groups shows the largest depression of freezing point. The HBD with higher hydrogen bond strength has the lower freezing point of DES. Because the interaction between HBD and anion groups increases the molecular size of DESs and reduces the interaction with cationic groups, thereby lowering the freezing points. For example, the freezing points of DESs formed by choline bromide, choline nitrate, choline chloride, with urea in a molar ratio of 1:2 are 1°C, 4°C,12°C, respectively(Abbott et al. 2003). The freezing points of DESs derived from choline salt decrease in the order of $F^- > NO_3^- > Cl^-$, indicating that freezing points are related to the strength of the hydrogen bond(Abbott et al. 2003). F^- has the biggest electronegativity and the smallest atom radius, thus can form strongest hydrogen bond with HBA, resulting in the lowest freezing point. N and Cl atoms have the weaker ability to form hydrogen bond compared with F. On the other hand, the HBA /HBD molar ratio also has an important impact on the freezing temperature. When ChCl and urea were mixed in a molar ratio of 1:1 and 1:2, the freezing points of the resulting DESs were > 50 °C and 12 °C, respectively (Abbott et al. 2003). Appropriate increase in the content of urea provides more possibilities for hydrogen bonding with ChCl, resulting in a lower freezing point.

Since DESs have a relatively wide liquid temperature range and low vapor pressure, many reactions in the DESs are carried out at mild temperatures and atmospheric pressure (Liu et al. 2015). When the reaction temperature is high, the volatile and high vapor pressure characteristics of conventional organic solvents will lead to side reactions and low selectivity. On the other hand, a large number of volatile solvents increases the burden of separation. Due to the low or even negligible vapor pressure of DESs, they have been used in many chemistry fields, such as organic synthesis, ionic thermal synthesis, and polymer reaction. In organic reactions, DESs not only act as solvents for reactants but also as catalysts. For example, ChCl-Urea DES was used as a catalyst and reaction medium to produce the 1-aminoanthra-9,10-quinone derivatives, avoiding the use of volatile organic solvents and concentrated acids (Phadtare and Shankarling 2010). Sonawane et al. synthesized novel diphenylamine-based fluorescent styryl colorants by Knoevenagel condensation in ChCl-Urea DES (Sonawane et al. 2010). Singh et al. reported that DESs were effective in the selective N-alkylation reactions of aromatic primary amines, in which DESs were both catalyst and solvent (Singh et al. 2011).

DESs are widely being exploited in ionothermal synthesis. Due to the negligible vapor pressure of DESs, it can eliminate

Table 2 Freezing point (T_f) of the reported DESs and the melting point (T_m) of pure HBA or HBD

HBA		HBD		Ratio HBA: HBD	DES T_f or T_g/°C	Ref.
Species	T_m /°C	Species	T_m /°C			
ChCl	303	urea	134	1:2	12	(Shahbaz et al. 2011)
		propanoic acid	−21.5	1:2	< −80[a]	(Lu et al. 2015)
		phenylacetic acid	76.5	1:2	25	(Florindo et al. 2014)
		malonic acid	135	1:1	10	(Florindo et al. 2014)
		glutaric acid	132	1:1	–16	(García et al. 2015b)
		glycolic acid	10	1:1	–16	(Florindo et al. 2014)
		levulinic acid	33	1:3	–11	(Berthod et al. 2018)
		oxalic acid	190	1:1	34	(Abbott et al. 2004)
		malic acid	135	1:1	–56	(Florindo et al. 2014)
		glycerol	17.8	1:2	–40	(Abbott et al. 2007c)
		furfuryl alcohol	−29	1:3	–36	(Berthod et al. 2018)
		d-sorbitol	98	1:1	9	(Abbott et al. 2004)
		fructose	103	2:1	10	(García et al. 2015b)
		glucose	146	2:1	15	(García et al. 2015b)
		o-cresol	30	1:3	–24	(García et al. 2015b)
		adipic acid	153	1:1	85	(Abbott et al. 2004)
		benzoic acid	122	1:1	95	(Abbott et al. 2004)
		citric acid	149	1:1	69	(Abbott et al. 2004)
		succinic acid	185	1:1	71	(Abbott et al. 2004)
proline	228	oxalic acid	190	1:1	–14.5[a]	(Domańska et al. 2015)
		glycolic acid	10	1:1	–30.7[a]	(Francisco et al. 2013)
		malic acid	135	1:3	–44.4[a]	(Domańska et al. 2015)
betaine	293	urea	134	1:2	42.5[a]	(Germani et al. 2017)
		ethylene glycol	−12.9	1:4	< –60[a]	(Taysun et al. 2017)
		phenol	41	1:3	< –80[a]	(Sarmad et al. 2016)
		oxalic acid	190	1:1	–17.2[a]	(Domańska et al. 2015)
		mandelic acid	119	1:1	13[a]	(Zhou and Liu 2014)
		malic acid	135	1:3	–20.0[a]	(Domańska et al. 2015)
		levulinic acid	33	1:2	< 30[a]	(Van Den Bruinhorst et al. 2018)
menthol	44	acetic acid	16.6	1:1	–7.8[a]	(Sarmad et al. 2016)
glucose	146	citric acid	149	1:1	9.8[a]	(Dietz et al. 2017)
		tartaric acid	173	1:1	–18.3[a]	(Dietz et al. 2017)
sucrose	186	citric acid	149	1:1	–14.0[a]	(Dietz et al. 2017)

[a] Glass transition temperature (T_g)

the potential safety hazards of solvents in ionothermal synthesis under high pressure, thus making the synthesis of new materials in DESs safer (Liu et al. 2010). In 2009, Morris group claimed that a borate-containing material was prepared by ionothermal synthesis using ChCl-1,3-dimethylurea DES (Lin et al. 2009). In 2010, Liu et al. reported an ionothermal synthesis of layered zirconium phosphate in the tetrapropylammonium bromide (TPABr)-oxalic acid DES (Liu et al. 2010).

As a medium for monomer dissolution, DESs exhibit excellent performance in polymerization and polymer reactions compared with the conventional organic solvents and ILs (Tang and Row 2013). Most ILs can dissolve monomer polymers at high temperatures around 150 °C, while DESs can prepare elastomer at a relatively low temperature (Bica et al. 2011). For example, Serrano et al. prepared DES based on mixtures of 1,8-octanediol and lidocaine for the synthesis of

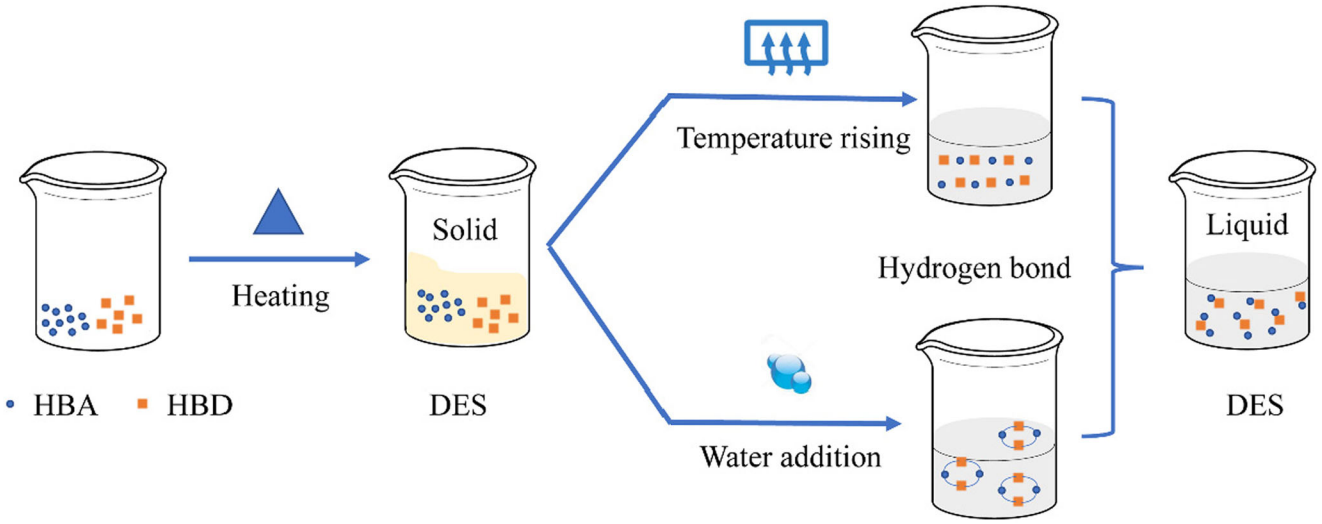


Fig. 2 Temperature and moisture enlarge the liquid range of DESs

poly(octanediol-co-citrate) elastomers at temperatures below 100 °C (Serrano et al. 2012). Compared with conventional organic solvents, this DES had better polymerization performance. Poly(acrylic acid)-carbon nanotube composites were synthesized by frontal polymerization in ChCl-acrylic acid DESs (Mota-Morales et al. 2013). Gutiérrez et al. claimed that ChCl-acrylic acid or ChCl-methacrylic acid DESs were superior to organic solvents and ILs in frontal polymerization (Gutiérrez et al. 2010).

Although DESs have lower freezing points than those of their constituents HBA and HBD, some of them such as ChCl: glucose/sucrose/fructose still have freezing points above 80 °C and remain as solids at ambient temperature (Vigier et al. 2015). To be used as extractive solvents, DESs should present as liquids at the extraction temperature, and the high freezing points limit their applications (Makoś et al. 2018). Since it is necessary to accurately understand the characteristics of DESs for a variety of industrial applications, the freezing points of DESs should be evaluated when designing solutions. A low freezing point broadens the operating temperature range and is conducive to the transmission of the intermediary substance during the separation process. Therefore, DESs with low freezing points are more favored by researchers. For example, the preferred DESs for biomass processing applications should possess freezing points below 50 °C. They can remain as liquids for better mixing and mass transfer efficiency (Satlewal et al. 2018). The freezing points of HBA and HBD are essential reference properties when we design DESs. Generally, the lower the freezing point of the counterparts, the deeper the decrease in the freezing point of the prepared DES. Studies about compositions, hydrogen bond strength, and freezing points of HBA and HBD provide the basis for the development of tailor-made DES for specific applications.

Surface tension

Although interfacial properties are essential for understanding mass transfer, few studies have well described the surface tension values of DESs. The surface tension of solvents is determined by the molecular cohesive tension. It can be measured by the platinum plate method with the surface tensiometer. Studies have shown that the surface tension of DESs is higher than those of molecular solvents and similar to those of imidazolium-based ILs (Zhang et al. 2012). Table 3 lists the surface tensions of DESs at 298.15 K and their temperature coefficients. The main factors affecting the DESs surface tension are the hydrogen bonding between mixture components, cationic type, and molecular weight. The hydroxyl group in the cation leads to higher surface tension due to their stronger hydrogen bonding ability. For instance, the surface tension of glucose-based DESs is higher than those of carboxylic acid-based DESs because they have more hydroxyl groups (Satlewal et al. 2018). Ghaedi et al. prepared six different DESs and found that the higher molecular weight caused the higher surface tension (Ghaedi et al. 2017b). The molar ratio of the HBA and HBD significantly impacts the surface tension of DESs. For example, the higher the molar ratio of ChCl: glycerol is, the lower the surface tension of ChCl-glycerol DES is. This phenomenon is due to the higher concentration of ChCl weakens the interaction between the HBA and the HBD (Zhang et al. 2012).

Surface tension values diminish greatly with increasing temperatures because of the increase of molecular kinetic energy and the decrease of the cohesion within the molecules, which weakens the intermolecular interaction. Abbott et al. measured the surface tension of ChCl-glycerol DESs. A linear correlation between surface tension and temperature was observed (Abbott et al. 2011). The surface tension of DESs fits a model according to the Eq. (1):

Table 3 Surface tensions of DESs at 298.15 K and their temperature coefficients

HBA	HBD	Ratio of HBA: HBD	γ (mN·m^{-1})	dγ /dT (mN·m^{-1} K^{-1})	Ref.
ChCl	Urea	1:2	52.0		(D'Agostino et al. 2011)
	Ethylene glycol	1:2	48.91	−0.0932	(Mjalli et al. 2014b)
	Malonic acid	1:1	65.98		(Abbott et al. 2004)
	Ethanolamine	1:7	49.18	−0.0694	(Mjalli et al. 2017)
	Glucose	2:1	71.71	−0.0516	(Mjalli et al. 2014b)
	Fructose	2:1	74.0	−0.200	(Hayyan et al. 2012)
Diethylethanolammonium^{+}Cl^{-}	Ethylene glycol	1:4	47.51	−0.0956	(Mjalli et al. 2014b)
	Glycerol	1:5	59.35	−0.1132	(Mjalli et al. 2014b)
Tetrabutylammonium^{+}Cl^{-}	Ethylene glycol	1:3	40.49	−0.0817	(Mjalli et al. 2014a)
	Glycerol	1:5	47.35	−0.0700	(Mjalli et al. 2014a)
	Triethylene glycol	3:1	40.22	−0.0849	(Mjalli et al. 2014a)
	Serine	8:1	68.34	−0.0103	(Mjalli et al. 2016)
	Threonine	9:1	40.60	−0.1050	(Mjalli et al. 2016)
	Methionine	11:1	39.89	−0.1070	(Mjalli et al. 2016)

$$\gamma = a + bT \quad (1)$$

where γ is the surface tension mN·m^{-1}. a and b are constants, and T is the absolute temperature. In addition, an increase in water content leads to a decrease in surface tension. For instance, in samples containing lactic acid and glucose, the surface tension of lactic acid-glucose (5:1) and lactic acid-glucose-water (5:1:3) are 37.42 and 35.01 mN·m^{-1}, respectively (Savi et al. 2019a).

Surface tension is one of the most important physical properties for DESs and many industrial processes involving various unit operations, such as mixing, fluid flow, and separation, which requires accurate surface tension values in unit design and equipment manufacturing. Surface tension can affect the wettability and contact of the packings in the gas-liquid reactor. In our previous study, we examined the surface tension of the 1,8-diazabicyclo [5.4.0] undec-7-ene (DBU) and glycerol DES system (DBU as HBA and glycerol as HBD), thus providing the basic surface tension data for designing and simulating absorption towers (Liu et al. 2018c). In the field of soil remediation, DES can be used to remove heavy metals in the soil. As shown in Fig. 3, Gupta and his colleagues proposed a green method to remove lead from contaminated soil. They used ChCl and glycerin mixing with

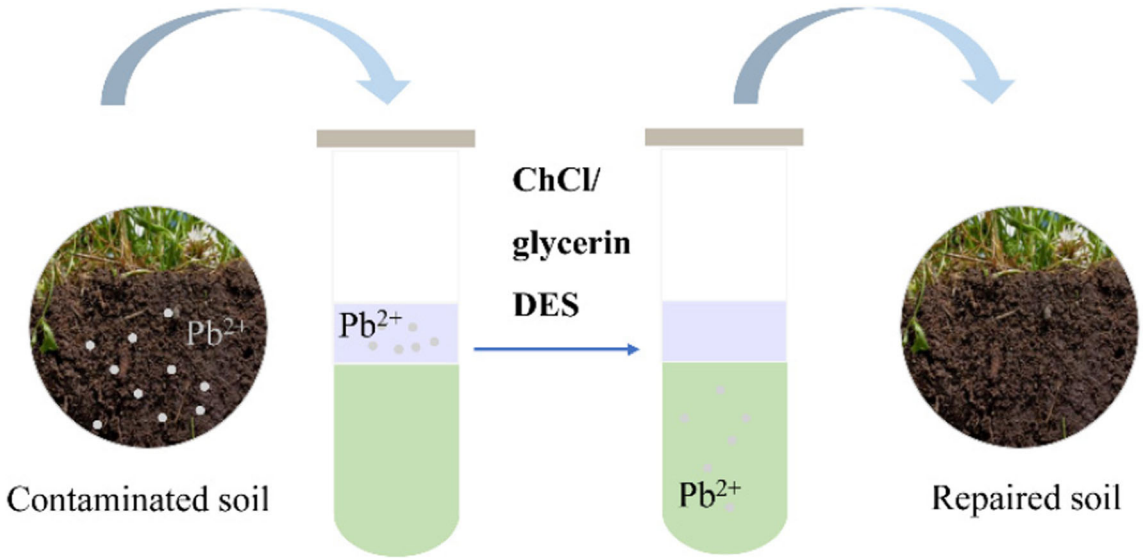


Fig. 3 Removal of heavy metals in the soil according to the surface tensions properties of DESs

the natural surfactant saponin for the removal of Pb^{2+} from the soil (Mukhopadhyay et al. 2016). In addition, malonic acid-based DESs altered the wettability of sandstone rock surface from liquid wettability to medium air-wetting conditions at the oil-air-rock system. They prevented the impact of water shock damage and stabilized the clays (Mohsenzadeh et al. 2015a; Mohsenzadeh et al. 2015b).

Surface tension has also received extensive attention in the emulsion and surfactant areas. The reaction in the two-phase mixture is often limited by mass transfer. DESs can reduce the surface tension of the water phase and act as surfactants, thereby improving the reaction activity in the aqueous phase. This ability of DESs is recently applied for the chemo-enzymatic epoxidation of soybean oil in an oil-water mixture (Lan et al. 2017). The conversion rate can be improved by adding various DESs, such as ChCl-Urea, ChCl-glycerol, ChCl-xylitol, and ChCl-sorbitol. The highest conversion rate is obtained in the ChCl-sorbitol system, far beyond the improvements by adding conventional surfactants (Lan et al. 2017). DESs containing of organic salt (HBD) and quaternary ammonium (HBA) salt with one or more long alkyl chains may exhibit surfactant properties. Chen et al. prepared polymer-based DESs acted as cationic and nonionic surfactants in a novel aqueous micellar two-phase system. Due to the surfactant characteristics and inherent charge, it exhibited a micellar-rich extraction environment (Chen et al. 2019).

Surface tension determines the suitability of liquids in mass transfer processes such as emulsification or separation. It provides important information about the influence of molecules on the strength of interactions in DESs (Ghaedi et al. 2017b). Higher surface tension makes it more difficult to penetrate the liquid surface and solution, which seriously affects the extraction efficiency. DES_S with lower surface tension may have better wettability, thus improving gas-liquid contact for gas absorption or pesticide preparation (Florindo et al. 2017a). Changing the hydrogen bond strength of HBA and HBD can adjust the surface tension to meet the application requirements of various mass transfer processes (such as distillation, absorption, separation, and extraction) (Savi et al. 2019a).

Density

Density is the basic physical property for chemical processing. It can be measured by the means of a specific gravity meter. Table 4 lists some density data of common DESs, and most of DESs have higher densities than water. Densities of DESs vary with the properties of individual component and their mole ratios. Anionic type significantly impacts the densities of solvents. For example, bromine salt DESs are denser than chlorine salt synthesis (García et al. 2015a). In the case of DESs formed by hydroxyl HBD, the densities increase with the number of hydroxyl groups (e.g., the values containing glycerol are greater than those containing ethylene glycol), but decrease with the introduction of aromatic groups (e.g., phenolic DESs have a lower density) (García et al. 2015a). Densities also decrease with the increase of the alkyl chain length of the DESs component (García et al. 2015a). In DESs containing diacid, the order of densities is oxalic acid > malonic acid > glutaric acid. Different molar ratios also affect density (García et al. 2015a). For example, the addition

Table 4 Densities of DESs at 298.15 K and their isobaric thermal expansibilities

HBA	HBD	Ratio	ρ /g·cm^{-3}	$10^3\alpha_p$ /K^{-1}	Ref
ChCl	Urea	1:2	1.198	0.482	(Liu et al. 2017)
	Triethyleneglycol	1:2	1.128	0.565	(Mjalli and Ahmad 2017)
	Glycerol	1:2	1.191	0.463	(Jibril et al. 2014)
	Furfuryl alcohol	1:3	1.135	0.581	(Lu et al. 2015)
	Fructose	2:1	1.118	0.527	(Hayyan et al. 2012)
	Glucose	2:3	1.267	0.407	(Wahab and Mahiuddin 2004)
	Mannose	1:1	1.278		(Florindo et al. 2017c)
	Ribose	1:1	1.267		(Florindo et al. 2017c)
	Xylose	1:1	1.257		(Florindo et al. 2017c)
	Levulinic acid	1:2	1.138	0.578	(Ghaedi et al. 2018)
	Oxalic acid	1:1	1.300	1.460	(Bahadori et al. 2013)
	Guaiacol	1:3	1.150	0.555	(Liu et al. 2017)
Choline bromide	Levulinic acid	1:4	1.207		(Wang et al. 2016)
Dodecyldimethyl-N-oxide	Phenylacetic acid	1:1	1.156		(Germani et al. 2017)
Octadecyldimethyl-N-oxide	Phenylacetic acid	1:1	0.977		(Germani et al. 2017)
N-dodecylmorpholinium-N-oxide	Phenylacetic acid	1:1	0.995		(Germani et al. 2017)

of ChCl to glycerol leads to the decrease of the density. Because of the higher the relative proportion of glycerol, the higher the structuring effect on the liquid and the lower the free volume of the solvent (Kareem et al. 2010).

As the temperature rises, the density decreases. This is because heating leads to more molecular activity and mobility, which increases the molar volume of DESs and eventually reduces density (Lin et al. 2018). It is well-known that density has a linear temperature dependence function:

$$\rho = a + bT \tag{2}$$

where ρ represents the experimental density of DESs, a and b are related constants (Hayyan et al. 2012). Considering the influence of temperature, derivation of density versus temperature to calculate its isobaric expansibilities (α_p):

$$\alpha_p = \rho^{-1}\left(\frac{\partial \rho}{\partial T}\right)_p = b/\rho \tag{3}$$

As shown in Table 4, densities are mostly within the range 1.1-1.4 g·cm^{-3}, and the isobaric expansibilities are within the range -0.55 to -0.65 K^{-1}. The density of nonionic DESs based on N-oxide for HBD may be less than 1. Hole theory is considered to explain this phenomenon (Abbott et al. 2007a). This notable difference in density may be attributed to the different molecular holes or empty vacancies. When the temperature is low, the average size of the pores is small, therefore reducing the molar volume of DESs and increasing the density. Apart from this, water content has a significant effect on the solvents' density. For example, Dai et al. found that densities of ChCl-glucose DESs decreased linearly with increasing water content (Dai et al. 2015). However, Florindo et al. tested the effect of water on five ChCl-based DESs (HBD: oxalic acid/glutaric acid/levulinic acid/glycolic acid/malonic acid). They found that the moisture content had no obvious influence on the density of DESs (Florindo et al. 2014).

The density of DESs is an essential attribute influencing liquid-liquid extraction because it determines whether the extractant collected from the upper phase or the lower phase. Figure 4 depicts the extraction process based on DESs density. Generally, hydrophobic DESs have a lower density than water. When such solvents are applied to liquid-liquid extraction, the extract-rich DESs phases are obtained from the upper liquid phase. In the extraction and separation, solvents used in the extraction process depends on their physical properties. The larger density difference between the selected solvents and the matrix is helpful for the phase separation in the liquid-liquid extraction technology (Huddleston et al. 1998). Density data is of great importance from the view of industrial applications. It can be used to evaluate the storage containers, determine overflow, and calculate column diameters during distillation, absorption, etc. (Ghaedi et al. 2017a). Density also provides important information about sample purity and molecular interactions in the liquid (Shahbaz et al. 2012). Compared with the traditional solvents, higher density DESs may be a problem in industrial-scale or continuous flow reaction (Ruesgas-Ramon et al. 2017). The density of DES can be fine-tuned through designing structures to suit the different chemical processes, reactors, and biochemistry, and so on.

Viscosity and electrical conductivity

Viscosity (η) is the key transport parameter of DESs, which describes the internal resistance of a fluid to shear stress(Ferraro 1956). It is usually measured by a viscometer. Table 5 lists the viscosity data of common DESs at 298.15 K. The viscosities of a large number of DESs are mostly in the range of 100 to 1000 mPa·s, which are higher than the viscosities of organic solvents. Figure 5 shows the viscosity relationship of ChCl-based DESs with different HBD components. The wide range of viscosity is usually attributed to the

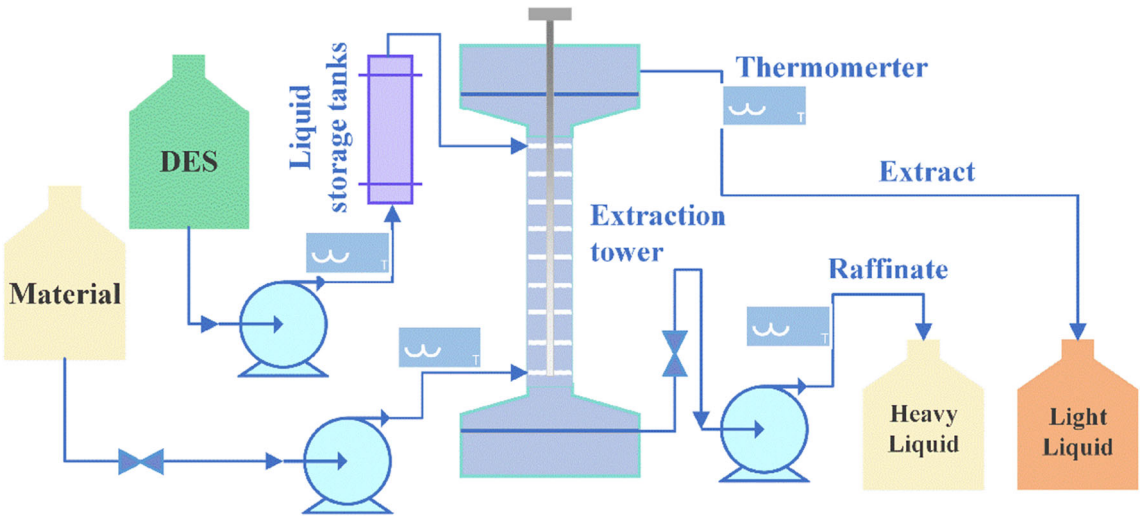


Fig. 4 Extraction process based on DESs density

Table 5 The viscosity of DESs at 298.15 K and parameters for the VFT correlation expression

HBA	HBD	Ratio	η /mPa·s	E_η /kJ·mol^{-1}	Ref
ChCl	Urea	1:2	829		(Abbott et al. 2016)
	Ethylene glycol	1:2	44.4		(Harifi-Mood and Buchner 2017)
	Glycerol	1:2	302	37.12	(AlOmar et al. 2016c)
	Triethylene glycol	1:2	839	40.31	(Bahadori et al. 2013)
	Phenol	1:3	446		(Guo et al. 2013)
	p-cresol	1:2	102		(Gajardo-Parra et al. 2019)
	Xylose	1:1	920		(Florindo et al. 2017c)
	Ribose	1:1	1564		(Florindo et al. 2017c)
	Glucose	3:2	7968	75.59	(Hayyan et al. 2013)
	Fructose	1:1	653		(Florindo et al. 2017c)
	Mannose	1:1	1068		(Florindo et al. 2017c)
	Xylitol	1:1	7540	65.86	(Maugeri and Domínguez De María 2012)
	Sorbitol	1:12	19470	70.91	(Maugeri and Domínguez De María 2012)
	Sucrose	1:1	79590	104.50	(Aroso et al. 2017)
	Lactic acid	1:1	400		(Altamash et al. 2017)
	Malonic acid	1:1	1350	46.66	(Sarmad et al. 2016)
	Levulinic acid	1:2	102	39.93	(Li et al. 2016a)
	Citric acid	1:1	3852	96.15	(Aroso et al. 2017)
	Tartaric acid	1:1	2389	102.08	(Aroso et al. 2017)
AcetylcholineCl	Levulinic acid	1:2	115	48.82	(Li et al. 2016a)
Betaine	Levulinic acid	1:2	981		(Dietz et al. 2017)
	Lactic acid+2H_2O	2:1		49.64	(Castro et al. 2018)
	Malic acid+3H_2O	2:1		55.85	(Castro et al. 2018)
Thymol	Camphor	7:3	18.8		(Makoś et al. 2018)
	undecylenic acid	7:3	15.6		(Makoś et al. 2018)
	decanoic acid	3:2	13.0		(Makoś et al. 2018)
decanoic acid	menthol	1:2	27.67		(Van Osch et al. 2019)
Histidine	Lactic acid+0.55H_2O	1:9		55.01	(Castro et al. 2018)

hydrogen bond network between HBA and HBD. The stronger interaction may contribute to the high viscosity (Ghaedi et al. 2017b). DESs containing ethylene glycol, phenol, or levulinic acid have relatively low viscosities (< 500 mPa·s) (García et al. 2015a), while viscosities of DESs are based on sugar, glycerol, and polycarboxylic acid are relatively higher (Florindo et al. 2014). Viscosities of DESs synthesized from polyols as HBD are generally lower than those of DESs from carboxylic acids and amines. Changes in the molar ratio of the DESs correlated with the viscosity of DESs. For example, the viscosity of ChCl-ethylene glycol DES increases as ChCl is added to the mixture (Abbott et al. 2007d). However, for ChCl-imidazole, the viscosity decreases as the composition of imidazole increases (Deng et al. 2017). For DBU-glycerol systems, with the increase of DBU content, their viscosities first raise rapidly to the maximum value and then decrease gradually to the viscosity value of pure DBU (Liu et al. 2019). This phenomenon is probably due to the competition between HBA-HBD interactions (intermolecular hydrogen bonds and intermolecular forces) and the HBA-HBA or HBD-HBD interactions. When the interaction between DBU and glycerol molecules is stronger than those between glycerol molecules, the viscosity of the mixing systems increases gradually, otherwise it decreases. Viscosities are also related to the alkyl chain length. Longer alkyl chains have higher viscosity. In addition, larger anions (such as Br^-) have a higher viscosity than smaller anions (such as Cl^-) (Elgharbawy et al. 2020).

Electrical conductivity indicates the strength of conducting electricity of solvents. The usual way to measure solvent electrical conductivity is through a conductometer. It also provides information about the free ions in the solution and ion migration rates. Electrical conductivity (σ) is strongly correlated with viscosity. As the viscosity increases, conductivity

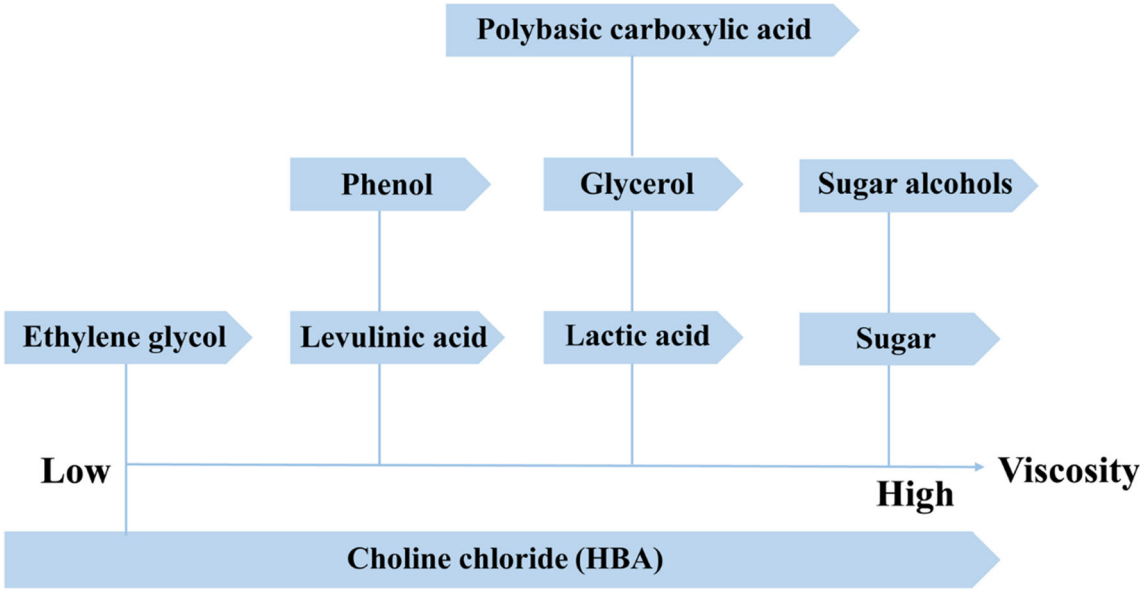


Fig. 5 Viscosity relationship of ChCl-based DESs with different HBD components

may decrease. Owing to the relatively high viscosities, most DESs have poor conductivity (lower than 2 $mS{\cdot}cm^{-1}$) (Hayyan et al. 2015). Table 6 lists some conductivity data of DESs at 298 K. The conductivity of DESs is affected by the number of charged substances and their mobility (Abbott et al. 2007d). Walden rule describes the relationship between viscosity and conductivity:

$$\Lambda\eta = k \tag{6}$$

where Λ represents molar conductivity, which is calculated from conductivity and density. k is the temperature-dependent constant. Λ represents the ion migration rate, and the derivative of viscosity φ ($\varphi=\eta^{-1}$) represents the fluidity with the surrounding medium. The curve of σ and φ changing with temperature displays an inverse linear relationship (Protsenko et al. 2017; Abbott et al. 2004;Protsenko et al. 2015). If a strong linear correlation exists for DESs, the conductivity is limited by ion mobility. Walden rule can qualitatively evaluate the ionicity. It is drawn on a $\log\Lambda$-$\log\varphi$ scale and compared with the so-called ideal reference line established by a 0.01 M KCl aqueous solution. The ideal line deviation indicates the interaction between anion and cation of solvents (Bahadori et al. 2013; Emi and Bockris 1970). Studies have found that DESs are more deviated from the ideal reference line than most ILs, suggesting the non-ionic structure of DESs and supporting the view that DESs cannot be regarded as ILs (Pinkert et al. 2011). DESs with low

Table 6 The electrical conductivity of some DESs at 298.15 K and parameters for the VFT correlation expression

HBA	HBD	Ratio	σ /$mS{\cdot}cm^{-1}$	E_σ /$kJ{\cdot}mol^{-1}$	Ref
ChCl	Glycerol	1:2	1.02	−26.92	(AlOmar et al. 2016c)
	Triethylene glycol	1:2	1.41		(Hayyan et al. 2015)
	Phenol	1:3	3.14		(Guo et al. 2013)
	Xylitol	1:1	1.15	-9.20	(Craveiro et al. 2016)
	Malonic acid	1:1	0.91	−30.96	(Bahadori et al. 2013)
	Levulinic acid	1:2	0.81	−5.99	(Li et al. 2016a)
	$Zn(NO_3)_2$ +$6H_2O$	1:1	9.28	−16.54	(Bahadori et al. 2013)
Acetylcholine Cl	Levulinic acid	1:2	0.67	−8.42	(Li et al. 2016a)
Betaine	Lactic acid+$2H_2O$	2:1		33.80	(Castro et al. 2018)
	Malic acid+$3H_2O$	2:1		35.77	(Castro et al. 2018)
Histidine	Lactic acid+$0.55H_2O$	1:9		36.07	(Castro et al. 2018)

viscosity exhibit lower ionicities, whereas very viscous ones lie closer to the ideal line (García et al. 2015a).

In general, viscosity and electrical conductivity are greatly affected by temperature. Viscosity diminishes greatly with increasing temperature, while electrical conductivity in reverse (Zhang et al. 2020a). Heating will increase the average velocity of the molecules in the liquid and reduce the internal resistance of the molecules. At the same time, the hydrogen bond between HBA and HBD weakens with the increase of temperature, leading to a decrease of mixture viscosity (Hayyan et al. 2012). The relationship between DESs viscosity and temperature is a nonlinear behavior and generally follows the Vogel-Fulcher-Tammann (VFT) expression:

$$\ln(\eta) = A\eta + B\eta/(T - T_0) \tag{4}$$

where η represents the experimental viscosity, T is the absolute temperature, $B_\eta = E_\eta / R$. E_η refers to the energy for the activation of viscous flow, which is closely related to the percentage of free volume in the liquid. T_0 is the fixed reference temperature, R is the molar gas constant, 8.314 $J \cdot mol^{-1}$ K^{-1}; The parameter A_η depends on the mole fraction of the HBA.

In many cases, the viscosity-temperature relation follows the Arrhenius expression:

$$\ln\eta = \ln\eta_0 + \frac{E_\eta}{RT} \tag{5}$$

That is, T_0 =0 in the VFT expression. The E_η of each system is calculated from the slope of $\ln\eta$ and $1/T$. It is used to quantify the temperature dependence of viscosity and the strength of intermolecular forces in DESs. Higher values are related to larger molecular sizes and stronger interactions. Low viscosity DESs (ChCl-glycol) show low E_η values, while high viscosity DESs (glycol-based) show larger E_η values (García et al. 2015a). The E_η values of the DBU-glycerol systems (the ratio of DBU is 0~1) are found to cover the range 23 to 79 $kJ \cdot mol^{-1}$, and the maximum values appear when the DBU ratio is about 0.6. This phenomenon is explained by the strong intermolecular hydrogen bond between glycerol and DBU under this composition, resulting in a decrease of porosity in the solution (Liu et al. 2018c). For DESs from ChCl or betaine with sugar molecules (glucose, xylose, and sucrose) and carboxylic acids (citric acid and tartaric acid) without water participation, the E_η values are about 100 $kJ \cdot mol^{-1}$ (Aroso et al. 2017).

Water molecules can act as plasticizer agents, disrupting networks of hydrogen bonds formed between DESs components, thus reducing the viscosity of the system (Lapeña et al. 2019). Introducing water in DESs causes a sharp decrease in viscosity. For instance, adding about 10 % water to the sample lactate-glucose (5:1) DES significantly reduced the viscosity of the system by 85.9 % (Savi et al. 2019a). Even if a small amount of water was added, the viscosities of DESs systems were also significantly reduced (Dai et al. 2013a; Savi et al. 2019b). However, the correlation between conductivity and water content in DESs is almost Gaussian (Liu et al. 2018b). Introducing a small amount of water into eutectic mixtures can result in their conductivity increment. More water addition causes the increase of the resistivity and dilution of the electrolytes, leading to a consequent decrease in conductivity.

Viscosity is an essential parameter in chemical, biological, and pharmaceutical applications. It also affects the efficiency of mass transfer and heat transfer and plays an important role in the chemical transformation of solutions (Yadav et al. 2014). Hydrophobic DESs with a low viscosity are normally used to extract nonpolar compounds. Ribeiro et al. prepared a series of hydrophobic DESs from menthol and fatty acids (Ribeiro et al. 2015). Due to the low viscosity caused by weak hydrogen bond interaction, they are beneficial to improve extraction efficiency. Li et al. synthesized six different molar ratios of ChCl-caffeic acid-formic acid DESs with lower viscosity and applied them to the molecularly imprinted polymers (MIP). They tested the absorption capacity of DESs for levofloxacin and discovered that they could further be used in drug purification and testing (Li and Row 2017). Olivares found that adding a small amount of water reduced the viscosity of betaine-urea DES, and even 2 % w/w of water would keep the solvent stable (Olivares et al. 2018). This DES can prevent the degradation of β-lactam antibiotics.

Viscosity is one of the essential factors in biomass processing. DESs with a relatively low viscosity show more substantial solubility for lignin. Zhao et al. found that ethanolamine-based DESs had lower viscosity causing stronger solute diffusibility in the pretreatment of wheat straw, which was conducive to the dissolution of lignin in DESs (Zhao et al. 2018). New et al. extracted palm leaves by mixing ChCl-urea DES with distilled water of 30 vol %. The enhanced delignification efficiency was attributed to the addition of water which reduced the viscosity of the DESs (New et al. 2019). Wahlström et al. increased the hydrolysis ability of xylose by adding a small amount of water to reduce the viscosity of the solvents. The furfural yield reached 44.87 % after adding 5 wt % water to the molar ratio of 1:3 ChCl and malic acid DES for 4 h at 130 °C (Morais et al. 2020).

Electrochemical application is an important field for DESs, and there is an urgent need for information about their electronic properties, especially conductivity. DESs could replace traditional organic solvents and catalysts to act as electrolytes and catalysts in electrochemically driven reactions. Water has a relatively narrow electric potential window, so the deposition of some metals is hampered by poor current efficiency. DESs have lower conductivity than ILs and excellent electrochemical stability, which can replace many substances as electrolyte solutions (Li et al. 2016c). Compared with other non-aqueous solvents, the main advantages of using DESs to replace aqueous electrolytes are that no electrochemical reaction

in a wide potential window and the high solubility of metal compounds including chlorides, oxides, and hydroxides. For example, Ramesh et al. developed polymer electrolytes utilizing lithium bis(trifluoromethanesulfonyl) imide (LiTFSI), cellulose acetate, and DESs. The rise of DESs content in polymer electrolytes can progressively enhance the ionic conductivity and electrochemical stability (Ramesh et al. 2013). The rise of DESs content in polymer electrolytes can progressively enhance the ionic conductivity and electrochemical stability (Ramesh et al. 2013).

At present, one of the main applications of DESs is to incorporate metal ions into DESs for metal deposition, dissolution, and processing (Almeida et al. 2020). DESs are suitable solvents for the testimony of zinc, copper, nickel, silver, gold, and tin. They also show potential for aluminum deposition and high-performance alloy electropolishing. The choice of HBD provides selectivity for extracting certain metals from complex matrices. Urea, thiourea, and oxalate are commonly used as HBDs (Smith 2013). For example, Abbott et al. electrolyzed nickel chloride dihydrate in ChCl-urea and ChCl-ethylene glycol to deposit Ni (Fig. 6) (Abbott et al. 2008). DESs also dissolve a variety of metal oxides, creating a "green" strategy for separating and recycling metals. Thicker metal films are more likely to be deposited in DESs, due to the high solubility of metal oxides and hydroxides in DESs rather than passivation (Smith 2013). The ChCl-$CrCl_3$ $6H_2O$ (1:2) DES was well suited for electrodeposition of thick and attached chromium films (Wright et al. 2012). Gu et al. used ChCl-urea DES to dissolve transition metal chlorides such as $NiCl_2{\cdot}6H_2O$, $CrCl_3{\cdot}6H_2O$, $FeCl_3$, and $CoCl_2{\cdot}6H_2O$ (Gu and Tu 2011). The ChCl-glycol/urea DESs were prepared for Zn-Sn alloy coating on low carbon steel substrates (Abbott et al. 2007b). Zinc and tin could be electrodeposited individually or as alloys from DES, and the selection of DESs could change the shape and composition of the alloy. In various ChCl-based DESs, silicon carbide, alumina, or diamond were stable over a long time. The electrolytic deposition of the metal from these liquids caused the physical binding of the particles to the metal coating to form a corrosion barrier that reduced surface friction and significantly improved metal hardness (Abbott et al. 2009).

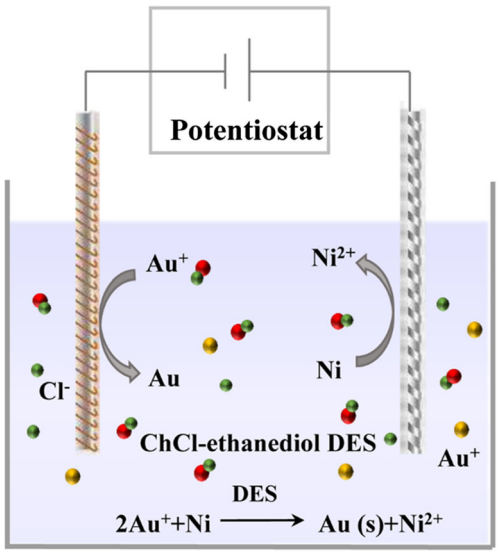


Fig. 6 Deposition of Ni in DES

The high viscosities and low conductivity of DESs may hinder their practical applications in different industrial processes. However, low viscosity also means weak molecular interactions that that weaken the decrease of the freezing point and the solubility. Therefore, the design of DESs viscosity should balance transmission performance and solvation properties. Taking into account the advantages of DESs' strong solvation ability in electrochemistry, exploring the full potential of their applications requires a deeper understanding of DESs in terms of composition and structure. The viscosity and conductivity of DESs can be customized by changing the structure, temperature, and water content. In the following research, the development and synthesis of novel DESs can be broadly implemented on an industrial scale.

Polarity

The polarity of DESs is an important parameter that determines their extraction capacity and miscibility with other solvents. The standard method to evaluate solvent polarity is through the solvatochromic parameters, which measures the blue-shift or red-shift of UV-vis bands. The solvents polarity parameters E_T *(NR)* and E_T *(30)* of Nile red and Reichardt's dye scale can be used to quantitatively assess the polarity of DESs(Liu et al. 2018b).

$$E_T\ (\mathrm{NR}) = hcv_{\max}N_A = 28591/\lambda_{\max,\mathrm{NR}} \tag{6}$$

$$E_T\ (30) = hcv_{\max}N_A = 28591/\lambda_{\max,30} \tag{7}$$

where NR stands for Nile Red dye, and 30 is Reichardt's dye. h stands for Planck's constant, c is the speed of light, and $\lambda_{\max}$ is the wavelength of maximum ultraviolet absorption. The $\lambda_{\max}$ value of DESs depends on the polarity of the medium, and a higher polarity medium results in a longer absorption wavelength. For example, the change of $\lambda_{\max}$ from water to pentane using Nile Red dye exceeds 100 nm (Deye et al. 1990). Pandey et al. measured the polarity of ChCl-glycerol/urea/ethylene glycol DESs with a molar ratio of 1:2 and confirmed that the polarity of DESs was higher than those of short-chain alcohols and common imidazolium and pyridinium ILs (Pandey et al. 2014).

Other indices, such as the Kamlet-Taft polarity parameter π^*, can also present the combined polarity of the solvents. π^* can be obtained by using the probes 4-nitroaniline and their $E_T(30)$ values (Florindo et al. 2017b). The solubility of the

substances in DESs are closely related to the Kamlet-Taft polarity parameter, and studies have shown that the DESs with high π^* values are more likely to dissolve lignin (Alvarez-Vasco et al. 2016).

Generally, polarity increases with increasing intermolecular attraction. It is mainly related to the molecular structure. Carbohydrate-based DESs present larger polarities compared with short-chain alcohols (e.g., ethanol, 2-propanol) and some polar aprotic solvents (e.g., dimethyl sulfoxide and dimethylformamide) (Ruß and König 2012). In addition, DESs composed of organic acids have the highest polarity, followed by sugar and amino acid-based DESs, while polyalcohol-based DESs are less polar (Dai et al. 2013a). The molar ratio of HBA and HBD also affects the polarity of DESs. A higher molar ratio of ChCl-glycerol increases the $E_T(30)$ value of DESs (Pandey et al. 2014). Abbott et al. studied the polarity of ChCl-glycerol DESs with different molar ratios and found that the polarity increased linearly with the increase of ChCl concentration (Abbott et al. 2011).

The polarity of DESs decreases significantly as the temperature increases due to the fracture of the hydrogen bond network (Pandey and Pandey 2014). Introducing water into eutectic mixtures also has an obvious effect on the polarity. The hydrogen bond interactions between DES components will begin to weaken during dilution and results in an increase of the polarity (Liu et al. 2018b). For example, Dai et al. assessed the polarity of different proportions of water on lactate-glucose and showed that 50 % (v /v) water content greatly influenced the polarity of DESs (Dai et al. 2013a).

They can dissolve many highly polar substances (Liu et al. 2018b). Polarity is usually used to predict the solubility of compounds in a solvent, but it is not always accurate because polarity is supposed to be described by multiple parameters. It is more appropriate to express solubility in terms of solvation capacity, which depends on electrostatic, inductive, dispersion, charge transfer, and hydrogen bond interactions. DESs with excellent solvability and a broad polarity range have greater dissolving capacities for various compounds. This would be efficiently applied in some fields, such as polymer dissolution (lignin, chitin, starch), drug solubilization, biodiesel synthesis, extraction, and gas absorption (Liu et al. 2018a; Francisco et al. 2012; Wagle et al. 2014).

Due to the selective solubilization, DESs have been applied to the wide area of delignification, cellulose extraction, and nanofibrillation. They are more commonly used to extract, separate, or purify some biopolymers, particularly the fractionation of lignocellulosic materials, which provides an economically advantageous and less toxic alternative to ILs (Wahlström et al. 2016). In terms of biomass separation applications, ChCl-lactic acid, betaine-lactic acid, or certain amino acid-based DESs (lactic acid-proline, malate-proline, malate-glycine) are useful in the dissolution of lignin and have a very low or negligible solubility in cellulose. Therefore, they have been applied for selective delignification of wheat straw (Francisco et al. 2012). Xu et al. prepared ChCl-formic acid DES for the pretreatment of corn stalks. They found that it could efficiently remove hemicellulose (66.2 %) and part of cellulose (Xu et al. 2016). Zhou et al. used a two-component system to dissolve cellulose, with the order of solubility of urea-acetamide < caprolactam-acetamide < urea-caprolactam (Zhou and Liu 2014). In the field of biomass conversion, DES systems lead to a sustainable pretreatment method in nanofibrillated cellulose (NFC) and cellulose nanocrystals (CNCs) production. DESs based on urea and ammonium thiocyanate or guanidine hydrochloride were used as novel green pretreatments for cellulose nanofibril production. Both DESs promoted the cellulose structure, loosening as indicated by an increase in fiber lateral dimension, while the degree of polymerization (DP) values of cellulose remained at the initial level after DES treatments (Li et al. 2017).

It has been reported that the eutectic formation of drug molecules through hydrogen bond interactions can improve the solubility of drug (Zhang et al. 2020b). Due to the extensive hydrogen bond network, DESs enhance the solubility and bioavailability of insoluble compounds in early drug development. For example, Morrison et al. measured the dissolution of several insoluble drugs such as benzoic acid, griseofulvin, danazol, and itraconazole in DESs. Compared with pure water, the solubility of these drugs in ChCl-urea and ChCl-malonic acid DESs was 5–22 times higher (Morrison et al. 2009).

DESs have a high solubility for polar compounds and low solubility for non-polar substances, so they can be used as an extractant to separate mixtures. The polarity of DESs is an important factor that affects the extraction efficiency. Dai et al. reported some DESs with different polarity, viscosity, and composition and found that DES composed of ChCl and 1, 2-propanediol had the lowest solubility for polar compound and better solubility for non-polar substance (Dai et al. 2013a). Moreover, ChCl-malic acid (1:1) DES could separate ethanol from n-hexane (Samarov et al. 2017). Quinoline was separated from toluene and heptane by extraction with methyl triphenyl bromide-glycol/glycerol (Naik et al. 2016). Using DESs extraction can also separate polar compounds from each other. For example, phenyl alcohol was isolated from acetic acid or propionate by extraction with glycerin acetate (Maugeri et al. 2012).

Water is the most common solvent in biocatalysis. A significant disadvantage of water as a solvent in biocatalysis conversion is its high polarity, which results in a low solubility of the substrate (Patzold et al. 2019). In addition, some adverse effects may occur in aqueous solvents, reducing the yield of target products. DESs with the polarity between water and organic solvents are used as substitutes for aqueous solvents in biocatalysis. They can improve enzyme activity, stability, and regioselectivity in the process of biotransformations

(Florindo et al. 2019). In the lipase-catalyzed synthesis of glycolipids, sugar-based DESs can be regarded as enzyme-compatible solvents in general. They can be used as a suitable medium for lipase-catalyzed transesterification of sugar and vinyl fatty acids (Patzold et al. 2019). In this reaction the sugars served as DES component and substrate at the same time. The benefit of using ChCl:sugar DESs is that an organic solvent can be avoided and solubility problems of the sugar in nonaqueous solvents can be ruled out.

The ionic character and high polarity of DESs make it less volatile when absorbing gases in the mixtures. As a consequence, they have great potential for gas separation and purification, chemical fixation, etc. (Hou et al. 2018). In carbon dioxide capture, alkanolamines are the main solutions to remove CO_2. However, solvents regeneration required high energy and bring about environmental disadvantages (Gray et al. 2005). In our previous research, DBU-glycerol DES was used as a CO_2 absorbent. We verified that DBU-glycerol had a higher circulating CO_2 load than monoethanolamine (MEA) aqueous solution (Zhu et al. 2020). In recent years, studies have found that DESs may become alternative absorbents of CO_2 for alkanolamines (Aissaoui et al. 2017). Li et al. measured the solubility of CO_2 in ChCl-Urea DES at different temperatures and pressures (up to 13 MPa) (Abbott et al. 2006a). Lu et al. determined the solubility of CO_2 in the six DESs containing ChCl-levulinic acid/furfuryl alcohol by isoosmotic saturation method, showing that levulinic acid-based DESs could capture CO_2 more efficiently than the furfuryl alcohol-based DESs (Lu et al. 2015). SO_2 is one of the dominant air pollutants and is mainly emitted from the burning of fossil fuels. Yang et al. synthesized a series of DESs from ChCl and glycerol and used them to quickly absorb SO_2 under different environmental conditions. Figure 7 shows the flowchart for the process of SO_2 absorption in the ChCl-glycerol DESs. They found that ChCl-glycerol DESs with a molar ratio of 1:1 showed the optimal SO_2 absorption capacity of 0.678 $g \cdot g^{-1}$ at conditions of 20.0 °C and 0.1 MPa. The SO_2 could be desorbed and DESs could be recovered under an N_2 flow at 50 °C (Yang et al. 2013). Guo et al. synthesized tetrabutylammonium bromide (TBAB)-caprolactam DESs to absorb SO_2 in a mixed gas (Guo et al. 2010). Their studies indicated that DES with a molar ratio of 1:1 had the highest SO_2 absorption capacity.

Understanding the polarity of DESs is useful for solving the problem of phase separation. However, there are relatively few studies on the polarity of DESs. The composition and structure of DESs affect the polarity of the solvents. In consequence, when designing DESs, HBD groups in the corresponding ions are selected and adjusted to fine-tune the polarity of DESs, to improve their dissolution characteristics and further expand applications in different fields.

Hydrophilicity/hydrophobicity

According to the affinity to water, DESs may be classified into hydrophilic DESs and hydrophobic DESs. Due to their inherent hydrogen bonding ability, DESs generally exhibit a high degree of hydrophilicity and are miscible with water. Conventional DESs, consisting of ChCl, glycerol, urea, and ethylene glycol, are all hydrophilic solvents. Hydrophobic DESs are mostly nonionic, based on menthol and carboxylic acid (acetic acid, pyruvic acid, lactic acid, or lauric acid) or a combination of tetraalkyl-quaternary-ammonium salt and capric acid (Ribeiro et al. 2015; Van Osch et al. 2015). The

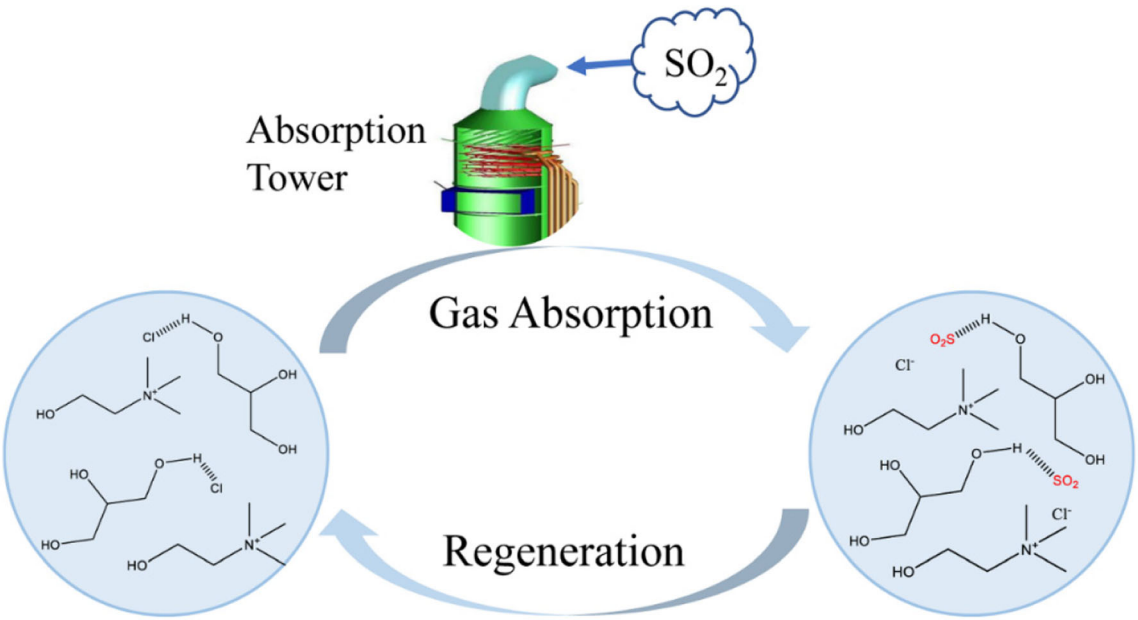


Fig. 7 Flowchart for the process of gas absorption in the ChCl-glycerol DES

hydrophilic and hydrophobic of DESs have a great influence on extraction and separation. Figure 8 shows the applications of DESs based on their hydrophobicity or hydrophilicity. Hydrophilic DESs are generally used for the extraction of polar and hydrophilic bioactive compounds, while hydrophobic DESs are more suitable for dissolving non-polar and hydrophobic compounds (Lee et al. 2019; Mbous et al. 2017).

Hydrophilic DESs are very effective in extracting natural bioactive compounds, such as alkaloids (Wang et al. 2017), flavonoids (Li et al. 2016b), xanthine (Machmudah et al. 2018), catechin (Jeong et al. 2017), and essential oils (Yu et al. 2017). They can also separate thiophene and benzothiophene containing sulfur compounds from biodiesel and other fuels. Khezeli et al. prepared two new hydrophilic magnetic DESs by mixing ChCl-phenol and ChCl-ethylene glycol with anhydrous $FeCl_3$. They are easily separated from the sample solution by applying a magnetic field, thus achieving a simpler and faster extraction of thiophene from heptane (Khezeli and Daneshfar 2017). Hydrophilic DESs have been used as a preconcentration medium to lower the detection limits of more sophisticated analytical techniques (Florindo et al. 2019). In microextraction, it can detect well-known micropollutants with low concentrations, such as pharmaceuticals, personal care products, metals, and dyes (Panhwar et al. 2018; Kaur et al. 2018). Smart materials such as magnetic gels, nanoparticles based on carbon nanotubes, and graphene have been prepared in DESs to remove arsenic, aluminum, lead, and mercury from water (Zarei et al. 2017; AlOmar et al. 2016a; AlOmar et al. 2016b).

In the two-phase system of extraction, it needs solvents that are immiscible with water (Cunha and Fernandes 2018). To achieve this, hydrophobic DESs are required, since they are immiscible with water and have high extraction efficiency for non-polar analytes. This type of DESs is recommended as potential extraction media to replace toxic organic solvents or expensive hydrophobic ILs (Lee et al. 2019). For example, several hydrophobic DESs made from fatty acids and quaternary ammonium salts are more suitable for the extraction of water-immiscible media in dilute aqueous solutions (Van Osch et al. 2015). Xu et al. synthesized four different hydrophobic DESs based on ChCl for protein extraction (Xu et al. 2014). Cao et al. developed the two-phase system using hydrophilic and hydrophobic DESs and simultaneously extracted various bioactive compounds with different polarity (Cao et al. 2018). In extractive separation, hydrophilic DESs are limited to solid or non-aqueous liquid samples. In contrast, hydrophobic DESs are usually used to extract various non-polar analytes from aqueous solutions. It is also possible to effectively extract various organic and inorganic analytes existing in the water environment into a variety of hydrophobic DESs and then are quantified using relevant analytical instruments.

Since hydrophobic DESs can extract nonpolar substances from aqueous solutions, many extraction processes are more effective by using these hydrophobic DESs (Cunha and Fernandes 2018). Tang et al. developed the headspace solid-single microextraction technique (HS-SME) to extract bioactive terpenoids by exposing ChCl-glycol DESs droplets with

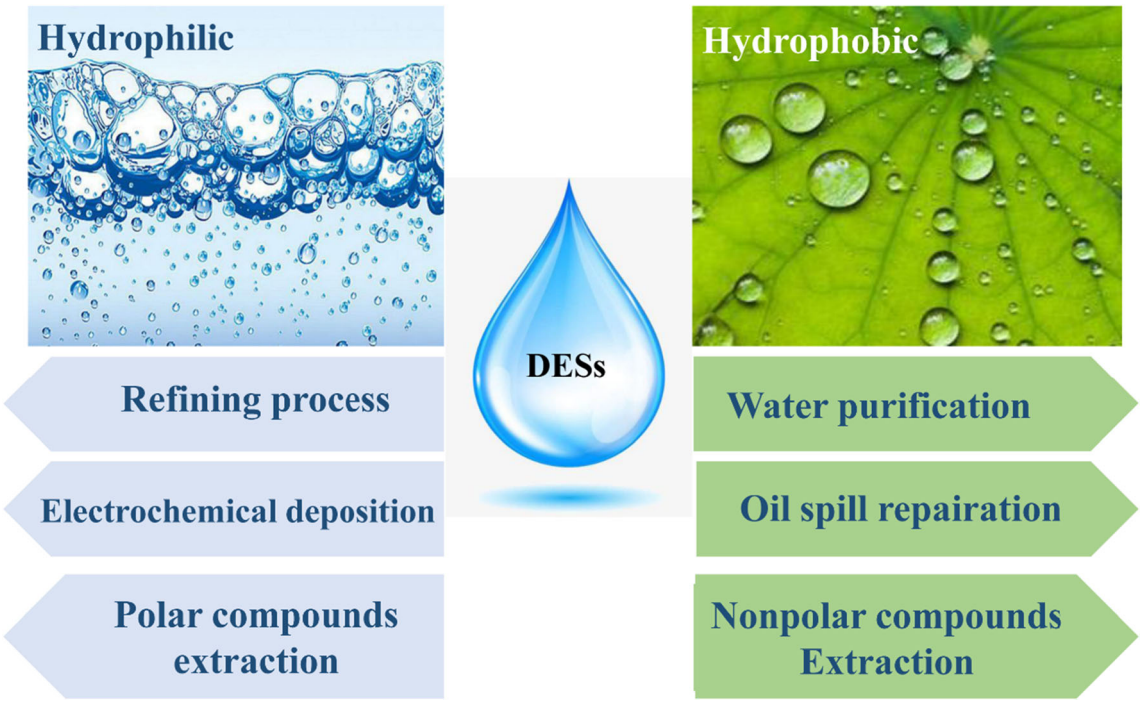


Fig. 8 DESs applications based on their hydrophobicity or hydrophilicity

the molar ratio of 1:4 to the headspace of leaf samples for 30 min (Tang et al. 2014). Gu et al. used a similar HS-SME procedure to extract phenolic compounds from crude oil (Gu et al. 2014). Zhuang et al. determined the flavonoid glycosides in the herbal medicine in ChCl-lactic acid DES after extraction (Zhuang et al. 2017). Moreover, the authors evaluated the possibility of recovering target analytes from DESs extracts to enable further applications in the pharmaceutical or food industries.

Interestingly, other hydrophobic DESs that do not contain ionic components also show significant extraction performance. For example, DL-menthol formed hydrophobic DESs with natural acids such as acetic acid, pyruvate, and lauric acid. They formed two phases with water and were used to extract biomolecules such as tryptophan, isophthalic acid, and vanillin (Ribeiro et al. 2015). Nonionic DESs formed from amide (N-methylacetamide) and long-chain carboxylic acid (lauric acid) could separate the oil phase in oil-water mixtures selectively and had broad application prospects in oil spill repair (Cui et al. 2017b).

Hydrophobic DESs can be used to remove micropollutants from water. When in contact with water, hydrophobic DESs facilitates the mass transfer between phases, thus achieving great efficiency in the removal of micropollutants. It is widely used to implement liquid-liquid extraction techniques to remove alkalis and transition metal ions from water. For example, Kroon et al. synthesized decanoic acid-lidocaine hydrophobic DESs to remove metal ions from non-buffered water (Van Osch et al. 2016). Menthol-based DESs could purify water by removing a variety of micropollutants from the water environment, such as pesticides (imidacloprid, Thiamethoxam, neonicotinoids, and acetamiprid) (Florindo et al. 2017a) and plastic additives (bisphenol A) (Florindo et al. 2018).

The practical applications of hydrophilic DESs are limited to polar compounds, and their high viscosity makes them difficult to phase separation. Hydrophobic DESs not only have obvious advantages in extraction and separation, but also are carriers for drug solubilization, and can replace many toxic organic solvents in analysis (Dwamena 2019). However, most research work mainly focused on hydrophilic DESs and relatively few studies focused on hydrophobic DESs. Up to now, the relationship between solvent structure and hydrophilic or hydrophobic application of DESs has not been found regularly. Since hydrophilicity and hydrophobicity are relative, the hydrophilic/hydrophobicity should be quantitatively evaluated in subsequent studies, such as the measurement of the oil-water distribution coefficient ($K_{o/w}$), etc. The applications of hydrophobic DESs in drug solubilization, biomass processing, analytical chemistry, and electrochemistry can be further developed.

Acidity/alkalinity

Another critical parameter for solvents performance of DESs is their acidity or alkalinity. This property can be measured by an instrument calibrated with a pH probe and an aqueous buffer solution. The Hammett function has been widely used to assess the acidity and alkalinity of solvents by determining the ionization ratio of indicators (Cui et al. 2017a). When a weak acid is selected as an indicator, the Hammett function parameter $H_$ is defined by the following equation.

$$H^{-} = \mathrm{pK}(HI) + \log([I^{-}]/[HI]) \tag{8}$$

where pK *(HI)* represents the thermodynamic ionization constant of the indicator in water. [I⁻] and [HI] are the molar concentrations of the anions and neutral indicators, respectively. The higher the $H_$ value, the more alkaline the medium. The acidity of highly acidic solvents is measured by their Hammett acidity function H_0 measured with an indicator (such as 4-nitroaniline).

$$H_0 = 0.99 + \log([I_{NA}]/[I_{NAH+}]) \tag{9}$$

where I_{NA} and I_{NAH}+ represent the light absorption intensity of the neutral and protonated forms of the indicator, the H_0 values of DESs, consisting of ChCl and HBD at mole ratios of 1:2, were reported as follows: p-toluene sulfonic acid 0.87, trichloroacetic acid 0.96, monochloroacetic acid 1.46, and propionic acid 2.23. The smaller the H_0 is, the more acidic a solvent is (Cui et al. 2017a).

Acidic DESs are easily formed when organic acids are HBDs. In salt-carboxylic acid based DESs, the acidity is mainly provided by the organic acid present in the mixture. The increase of the alkyl side chain of the compound reduces the solvent's ability to donate protons (Teles et al. 2017). On the other hand, pH decreases linearly with increasing temperature and water content (Hayyan et al. 2012). Quite a lot of DESs are highly acidic, such as ChCl-malonic acid, ChCl-citric acid, and ChCl-oxalic acid. Because of their good solubility, they are widely used in biomass conversion, extraction and separation, drug solubilization, and catalytic reaction. Acidic DESs have higher fractionation and delignification capabilities (Tan et al. 2018). For example, ChCl-carboxylic acid DESs preferentially dissolve lignin at 60 °C, thus can be used as useful solvents for delignification (Lynam et al. 2017). Furthermore, Biswas et al. found that 5–10% starch was found to be dissolved in the highly ChCl-oxalic acid DES. In contrast, DESs of the less acidic ChCl-succinate, ChCl-maleic acid, and ChCl-phenylacetic acid had poor starch solubility (Biswas et al. 2006). Morrison et al. found that ChCl-malonic acid based DES with higher acidity showed higher solubility of danazol, itraconazole, and griseofulvin (Morrison et al. 2009). In the dissolution of metal oxides, the stronger acidity of DESs, the better solubility of metal

oxides. For example, Fe_3O_4 was the most soluble in ChCl-oxalic acid but almost insoluble in ChCl-phenylpropionic acid. Malonic acid-based DESs have higher solubility for aluminates and silicates than urea and ethylene glycol-based DESs (Qin et al. 2019).

On the other hand, acidic DESs can promote acid-catalyzed reactions in many chemical productions. As a solvent or catalyst, acidity is critical for DESs, especially in esterification and transesterification (Qin et al. 2019). For example, a novel DES is prepared by mixing quaternary ammonium mesylate with p-toluene sulfonic acid (PTSA) (Qin et al. 2019). It is not only a solvent but also a catalyst for the esterification of carboxylic acid and alcohol. Qin et al. synthesized acid-base adjustable DESs by imidazole and PTSA to prepare long-chain organic esters (Qin et al. 2019). Generally speaking, DESs can be used as extractants in oxidative desulfurization and their physiochemical properties have obvious influence on the catalytic performance (Jiang et al. 2019; Mao et al. 2017a; Mao et al. 2017b). For example, Liu et al. synthesized four dual-acidic DESs in an oxidative desulfurization (ODS) system (Liu et al. 2020). These DESs demonstrated excellent properties for removing aromatic sulfur compounds. The sulfur removal rates were up to 99%. Furthermore, the desulfurization efficiency reached 98% after DES regeneration, achieved by simply washing with ultrapure water (Liu et al. 2020).

Neutral DES can be prepared when sugar, polyols, and metal salt are used as HBDs. Neutral DESs have a low sensitivity to temperature changes and are suitable for chemical, environmental and biological applications. Neutral DESs (such as ChCl-$ZnCl_2$) seem to facilitate the cleavage of the methoxy moiety on the lignin aromatic ring, which also leads to a significant increase in the phenolic hydroxyl group, leaving behind β-O-4 bond and little change in the molecular weight of treated lignin (Hong et al. 2016). In addition, ChCl-glycerol DES was widely used in the purification of biodiesel (Zhang et al. 2012). Zhu et al. evaluated the direct nucleophilic substitution reaction of alcohols. They found that ChCl: $ZnCl_2$ was used as both solvent and catalyst, achieving an excellent separation yield of 95% (Zhu et al. 2011). Azizi et al. found that ChCl:2$SnCl_2$ DES was used as a green and reusable catalyst for the electrophilic substitution reaction between indole derivatives and various aldehydes to synthesize diindole methane (Azizi and Manocheri 2012).

A few of the DESs are alkaline. The most commonly reported basic DESs include ChCl-urea, ChCl-triethanolamine, potassium carbonate-glycerol, etc. Alkaline DESs are beneficial to the dissolution of cellulose and have a relatively small degree of polymerization of carbohydrates and structural disintegration of cellulose (Mäki-Arvela et al. 2011). The alkalinity of DESs determines the solubility of cellulose. The stronger the basicity of DESs, the easier it is to break the hydrogen bond in supramolecular structure, thus making cellulose a higher solubility (Abe et al. 2010). Alkaline DESs are the catalysts for many base catalytic reactions, such as the Perkin reaction, Knoevenagel, and Diels-Alder reaction (Zhang et al. 2012). For example, Singh et al. found that ChCl-urea acted as both catalysts and solvents for the N-alkylation of aromatic amines. The ChCl-urea mixture was efficient in catalytic reaction, which might be due to its higher alkalinity (Singh et al. 2011). Alkaline DESs are the best choice for the separation and purification of acidic gases. Owing to the weak alkalinity of the ChCl–urea eutectic mixture, this DES is capable of absorbing a small amount of acid gas such as CO_2 (Zhang et al. 2012).

Figure 9 shows the possible applications of DESs according to their acidity or alkalinity nature. The acidity and alkalinity of DESs have an essential influence on the dissolution of compounds, catalytic reaction, and separation. However, there are few specific parameter data on DESs acidity and alkalinity, and relatively few studies on the influence of different structures on the acidity and alkalinity of solvents. The acidity and basicity of DESs will vary according to the nature and ratio of their counterparts. Based on the design of hydrogen bond acidity and alkalinity, the pH of solvents can be adjusted according to the specific applications. Thus, DESs can replace the diluted inorganic acid and expensive ILs in the physical and chemical process to meet the demand for green chemistry.

Stability

DESs have good chemical stability and thermal stability, and are not inflammable. Thermogravimetric analysis (TGA) is used to investigate the thermal stability of DESs. The higher decomposition temperature of DESs means that it can be used at a higher temperature and has better stability (Elgharbawy et al. 2020). The stability of DESs depends on their composition, molar ratio, and heating time. It is also affected by the number of HBD or HBA groups, spatial structure, and functional groups. DESs with a more carboxyl (-COOH) structure are more stable. The stability and formation of hydrogen bonds may be affected by the number of hydrogen bond groups and their three-dimensional structure (Elgharbawy et al. 2020). Simeonov measured the thermal stability of some common urea-based DESs (Simeonov and Afonso 2016). A eutectic mixture with carbohydrates and ChCl exhibited lower rates of decomposition. After heating for 7 h at 80 °C, some carbonate and ammonia emissions are detected, revealing the decomposition of DESs (Simeonov and Afonso 2016). Rengstl et al. studied the thermal stability of ChCl-based DESs and found that the decomposition temperatures of DESs were in the range of 269–280 °C. These temperatures are higher than those of HBD alone and similar choline ionic liquids (Rengstl et al. 2014). Zhao et al. examined that glycerine-based DESs had higher thermal stability (Zhao and Baker 2013). After studying the long-term isothermal

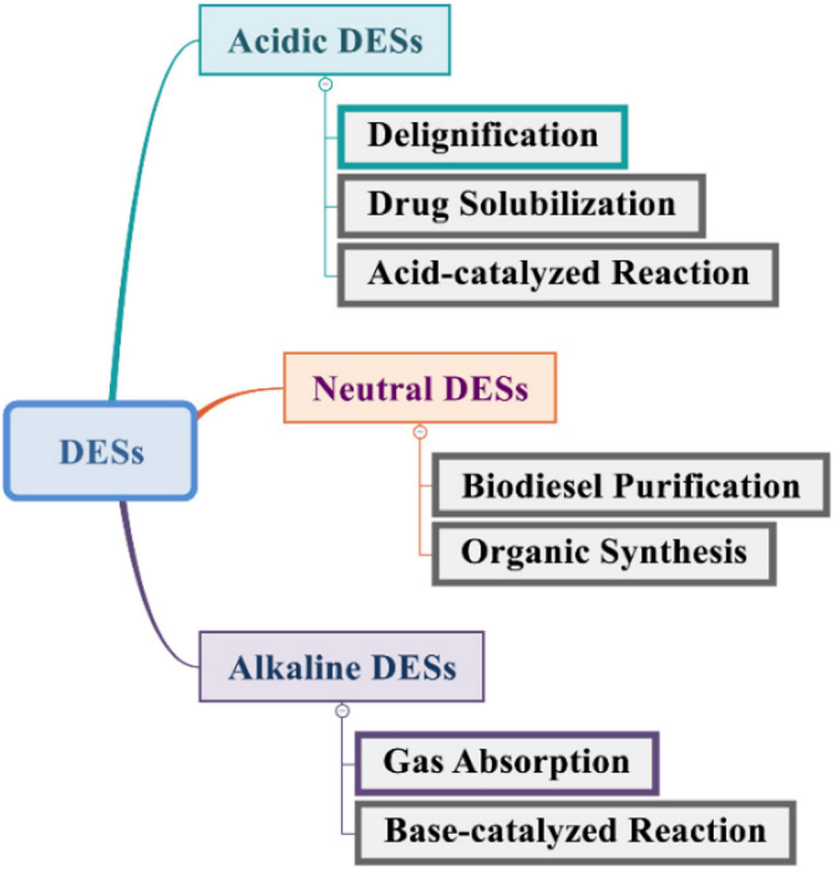


Fig. 9 Possible applications of DESs according to their acidity or alkalinity nature

stability of DESs, it is found that the order of thermal stability was ChCl-tartaric acid > ChCl-lactic acid > ChCl-malonic acid (Chen et al. 2018a). This is because the thermal stability of DESs depends on the number of hydrogen bonds provided by the acids and the thermal stability of the organic acid itself. Tartaric acid provides more hydrogen bonds and has good thermal stability, so the ChCl-tartaric acid DES has higher thermal stability.

Thermophysical properties of DESs, such as the temperature dependence of density and viscosity, are key parameters for the successful design and operation of the separation process (Elgharbawy et al. 2020). DESs have great potential in organic synthesis, chemical catalysis, and biomass conversion due to their high thermal stability and low vapor pressure, which enables reactions to take place at temperatures higher than 100 °C in non-pressure vessels (Huang et al. 2019). The thermal stability of DESs is critical when the processing temperature needs to be increased (e.g., dissolution, extraction, modification, and plasticization) (Zdanowicz et al. 2018). DESs act as plasticizers in the preparation of polysaccharides, such as thermoplastic starch or polymer blends with other thermoplastic polymers (e.g., biodegradable corn protein and polycaprolactone or non-biodegradable polyethylene) (Zdanowicz et al. 2018). Studies have shown that ChCl-based DESs are effective functional additives for thermoplastic starch, which can reduce the recrystallization tendency to the greatest extent. Leroy et al. investigated the decomposition temperature of ChCl-citric acid and ChCl-imidazole DESs, revealing that DESs with good thermal stability were more suitable for starch plasticization (Leroy et al. 2012). The considerable potential of DESs in formulation makes them very attractive for the development of effectively biodegradable plastics and polysaccharide composites.

The chemical stability of DESs is a prerequisite for their industrial applications and thermal stability, which provides a reference for their application temperature. DESs have strong hygroscopicity (especially when HBA is ChCl). Moisture content also affects the physicochemical properties and applications of DESs (Morais et al. 2020). Therefore, during storage and operation, understanding the distinct effects of water in the air on their stability is highly important. In the fields of organic synthesis, chemical catalysis, and biological transformation, the decomposition temperature of DESs should be considered first. The excellent stability of DESs makes it unnecessary to consider the troubles caused by the flammability of other solvents. Thus, it will be desirable to improve their safety and operability in industrial applications and reduce unnecessary investment.

Recycling and recovery systems

From the perspective of economy and recycling, the recovery and reuse of DESs as efficiently as possible are essential for industrial production. Due to the low vapor pressure and non-volatility, DESs are easy to separate and recover. However, they are, to some extent, miscible with water and inevitably occur in water-based environments, making their recovery even more challenging (Smith et al. 2014). The lifetime and recyclability of the DESs are regarded as key factors in many chemical processes and industrial applications. Shen et al. examined the effect of the recycled solvent on the digestibility and lignin structure characteristics of cellulose pulp. Throughout the four cycles, the pretreatment efficiency (delignification and enzymatic saccharification) could still maintain a high level. The mass balance analysis showed that the recovery yield of DESs was at least 90% and more than 80% delignification ratio was achieved (Shen et al. 2019).

The typical strategy approaches have been studied to recover DESs by the addition of antisolvents. DESs are at least partly miscible with protic solvents such as MeOH and EtOH, whereas immiscible with the aprotic solvents such as toluene, hexane, and ethyl acetate. These aprotic solvents can be used to recover DESs by liquid-liquid extraction. Figure 10 shows a flow chart of DES recovery in biomass transformation. Acetone can be used as an antisolvent to recover ChCl-urea DES from rice straw. Handy et al. first extracted with ether. Then, ChCl-urea and ChCl-glycerol DES were recovered by vacuum drying (Handy and Wright 2014). Phadtare et al. recovered and reused ChCl-urea DESs by first extracting with water and then rotating evaporation at 80 °C. The recovered DESs showed excellent recovery and reuse effect (Phadtare et al. 2013). Francisco et al. proved that acetone could be used as an antisolvent for the recovery of ChCl-lactic acid DESs, and delignification ratio was still largely maintained after recycling four cycles (Francisco et al. 2012). However, the adverse effects of dilution should also be considered when using antisolvents recovery DESs. For example, as lignin contains phenolic hydroxyl in its molecular structure when the added antisolvent is less than a certain amount, new DESs systems may be formed between lignin and ChCl in the original DESs. So that part of the lignin may be dissolved in the new DESs systems, resulting in a decrease of lignin recovery (Kim et al. 2018). Moreover, the accumulation of residual degradation products after the evaporation of the antisolvents might reduce the recovery performance of DESs.

The membrane processes such as nanofiltration, reverse osmosis, electrodialysis, and pervaporation have been investigated to recycle DESs. Haerens et al. used nanofiltration and reverse osmosis to recover ChCl-ethylene glycol DES. They found that osmotic pressure was the limiting factor in nanofiltration or change osmosis (Haerens et al. 2010). Liang et al. used electrodialysis to transfer the electrolyte components such as ChCl in DESs through the ion exchange membrane, while other components such as polyol were retained in the original part due to their non-electrolyte properties, thus realizing the effective separation and recycling of DESs (Liang et al. 2019).

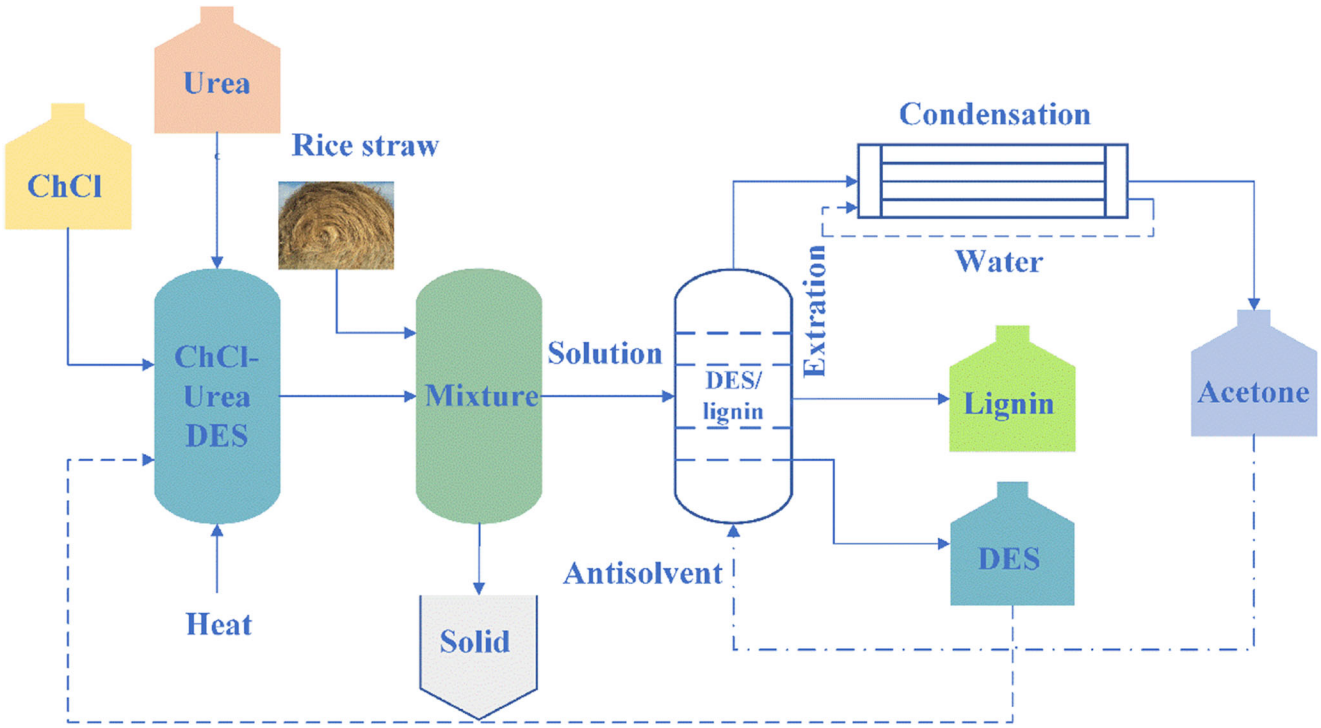


Fig. 10 Scheme depicting the strategy that can be adopted for the recovery of DESs

Generally, DESs are considered to be recycled more readily than many organic solvents. Their basic structure, physicochemical properties, and reaction conditions play important roles in how to recycle DESs (Chen et al. 2018b). It is necessary to develop efficient, low-cost, and green DESs recovery methods and in-depth study of their recycling efficiency in industrial applications.

Conclusion and future perspectives

Compared with common organic solvents and ILs, DESs provide an economical and green solvents selection strategy for chemical reaction and substance separation, which are simple to synthesis, low toxicity, biodegradable, and relatively unreactive with water. In addition, DESs exhibit excellent physicochemical properties and have gained extensive attention in many industrial applications. To optimize their solvent performance, DESs are easily tailored according to the structure, molar ratio, temperature, water content, and the addition of co-solvent. For specific applications, rationally design and synthesize DESs with different HBA and HBD components to have distinct physical and chemical properties, providing theoretical guidance for process design and large-scale industrial applications.

This review summarizes the properties of DESs and appropriate specific applications based on their different physicochemical properties, and systematically analyzes the relationship between DESs composition, structural performance, and application. In the past ten years, DESs have been used in organic synthesis, biomass catalytic conversion, extraction of bioactive compounds, gas absorption, extraction and separation, and have shown excellent performance and significant economic potential. Although various DESs for different applications have been developed extensively, further research is needed to create more functional DESs. First of all, the properties of DESs in the literature are not comprehensive enough, lacking data on physicochemical properties such as specific heat volume or absorption heat. Second, the relationship between the structure, composition, and physicochemical properties needs to be further improved to guide the design and development of solvents. Third, according to the characteristics of DESs, their functional applications should also be broadened, such as sewage treatment, protein crystallization, nano-drug preparation. Then, reveal the structural evolution and influence law of DESs in their applications and recycling process, and improve the recycling efficiency and green economy of DESs. More importantly, their applications are still in the laboratory stage, and their industrial applications development needs to be accelerated, such as DESs pulping process, gas desulfurization process, and polysaccharide processing technology. In the subsequent research, auxiliary molecular simulation and calculation methods are also the research direction, which can further systematically study the mechanism of DESs in different systems and clarify the mechanism. The precise guidance of DESs design and synthesis lays the theoretical foundation for a more efficient and green targeted application of DESs.

Author contribution All authors are involved in some work of the manuscript. Man Zhang was a major contributor in writing the manuscript. Xingyilong Zhang and Yingying Liu wrote the first draft of the main manuscript text. Kejing Wu and Yingming Zhu prepared the Figures and Tables. Houfang Lu and Bin Liang provided critical revision of the manuscript. All authors read and approved the final manuscript.

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Declarations

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Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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